

Tugas 12: Clustering Koordinat GPS Menggunakan Algoritma DBSCAN di Google Colab

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1. Pendahuluan

Pada praktikum ini dilakukan analisis data spasial menggunakan algoritma clustering **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**. Data yang digunakan berupa **koordinat GPS (latitude dan longitude)** dengan tujuan utama untuk **mendeteksi kelompok lokasi berdasarkan kepadatan titik**. Metode ini sesuai digunakan untuk data geolokasi karena dapat mengidentifikasi daerah padat, menentukan kluster tanpa harus menentukan jumlah kluster di awal, serta mampu **mendeteksi noise/outlier** secara otomatis.

2. Tujuan

Tujuan dari praktikum ini adalah:

- Mengolah data koordinat GPS menggunakan Python di Google Colab.
- Menjalankan algoritma DBSCAN untuk mengelompokkan titik-titik berdasarkan kemiripan lokasi.
- Menampilkan visualisasi hasil kluster dalam bentuk scatter plot peta dan diagram batang distribusi kluster.
- Memberikan penjelasan singkat untuk keperluan laporan analisis data.

3. Dataset

Dataset yang digunakan berisi data geografis dengan kolom koordinat:

- latitude
- longitude

The screenshot shows a Google Colab notebook titled 'Praktikum 12'. The code cell [1] defines three FILEPATH variables pointing to Google Drive links. The code cell [2] imports pandas as pd and reads the CSV file into a DataFrame df, then displays the first five rows of the dataset.

```
[1] FILEPATH = 'https://docs.google.com/uc?export=download&id=1Gy6kxau7zj0yp115Cha2yc7_fqId7qn'
FILEPATH = 'https://docs.google.com/uc?export=download&id=130517Pek113Jm6Q-t771yp70bHG1N1u'
FILEPATH = 'https://docs.google.com/uc?export=download&id=130517Pek113Jm6Q-t771yp70bHG1N1u'

[2] import pandas as pd
df = pd.read_csv(FILEPATH)
df.head()
```

	id	name	country_id	country_name	state_code	type	latitude	longitude
0	3901	Badakhshan	1	AF	Alghanistan	BDS	NaN	36.734772
1	3871	Badghis	1	AF	Alghanistan	BDG	NaN	35.167134
2	3875	Baghlan	1	AF	Alghanistan	BGL	NaN	36.178903
3	3884	Balkh	1	AF	Alghanistan	BAL	NaN	36.755060
4	3872	Bamyan	1	AF	Alghanistan	BAM	NaN	34.810007

4. Langkah – Langkah Praktikum

4.1 Memuat Data

Dataset dimuat menggunakan `pandas.read_csv()` dari Google Colab. Tujuannya untuk memastikan data berstruktur tabel dan dapat diolah.

4.2 Menyiapkan Koordinat GPS

Kolom latitude dan longitude diekstraksi dari dataset:

The screenshot shows a Google Colab notebook titled 'Ekstrak Koordinat GPS'. The code cell [3] extracts the 'latitude' and 'longitude' columns from the DataFrame df and prints the first five rows of the extracted coordinates.

```
[3] coordinates = df[['latitude', 'longitude']]
print("Extracted GPS Coordinates:")
print(coordinates.head())
```

	latitude	longitude
0	36.734772	70.811995
1	35.167134	63.769538
2	36.178903	68.745306
3	36.755060	66.897537
4	34.810007	67.821210

4.3 Menerapkan Algoritma DBSCAN

```
[6] ✓ Os # Adjust these values for optimal results based on data characteristics.
dbscan = DBSCAN(eps=0.05, min_samples=5)

# Fit the model and predict clusters
clusters = dbscan.fit_predict(scaled_coordinates)

# Add the cluster labels to the cleaned dataframe
df_cleaned['cluster'] = clusters

print("DBSCAN clustering applied to cleaned data.")
print("Number of clusters found (excluding noise):", len(set(clusters)) - (1 if -1 in clusters else 0))
print("First 5 rows of cleaned dataframe with cluster labels:")
print(df_cleaned[['latitude', 'longitude', 'cluster']].head())
```

DBSCAN clustering applied to cleaned data.
Number of clusters found (excluding noise): 5
First 5 rows of cleaned dataframe with cluster labels:

	latitude	longitude	cluster
0	36.734772	70.811995	0
1	35.167134	63.769538	0
2	36.178903	68.745306	0
3	36.755060	66.897537	0
4	34.810007	67.821210	0

 **Praktikum 12** ☆ 

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Subtask:

Menerapkan algoritma clustering DBSCAN pada koordinat GPS yang diekstrak. Parameter seperti 'eps' (radius lingkungan) dan 'min_samples' (jumlah sampel minimum) akan ditentukan. Perlu diingat bahwa nilai-nilai ini mungkin perlu disesuaikan untuk hasil optimal.

[4] ✓ Os

```
from sklearn.cluster import DBSCAN
from sklearn.preprocessing import MinMaxScaler
from sklearn.compose import make_column_transformer
print("Imported DBSCAN, MinMaxScaler, and make_column_transformer.")
```

Imported DBSCAN, MinMaxScaler, and make_column_transformer.

[6] ✓ Os

```
df_cleaned = df.dropna(subset=['latitude', 'longitude']).copy()
coordinates_cleaned = df_cleaned[['latitude', 'longitude']]

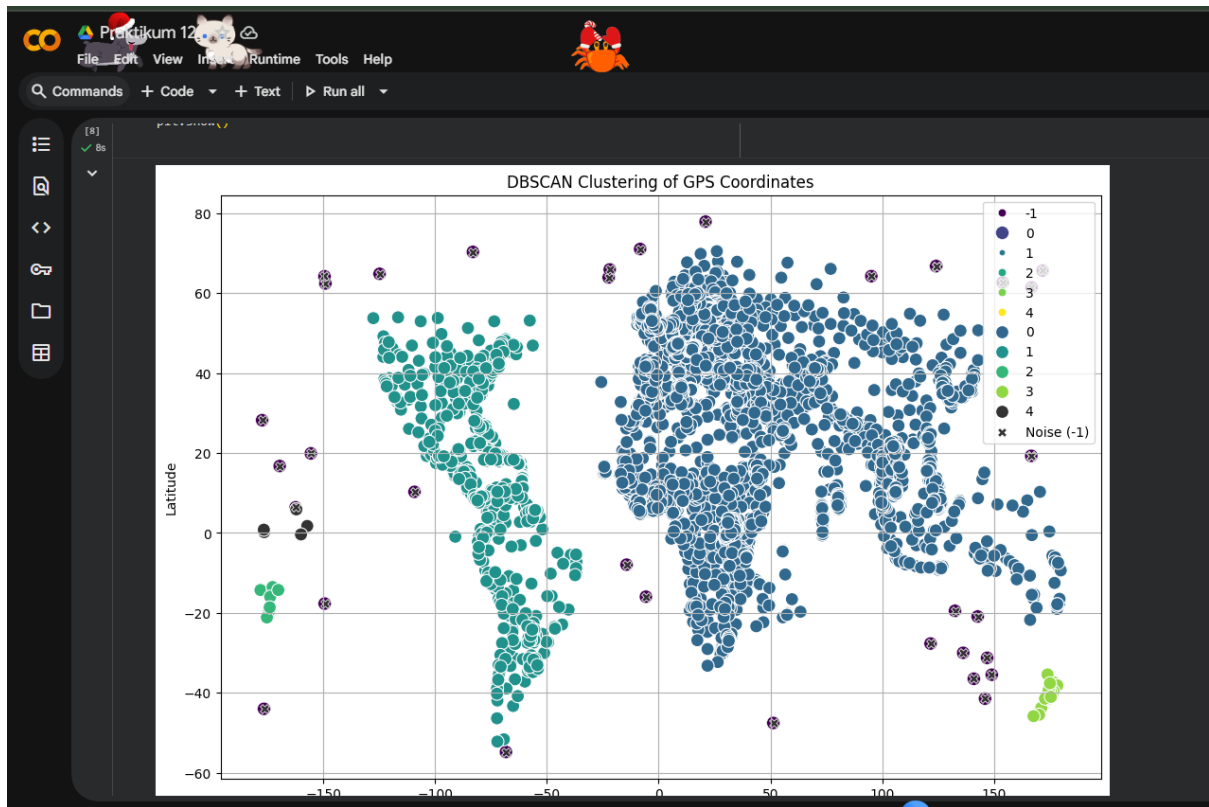
preprocessor = make_column_transformer(
    (MinMaxScaler(), ['latitude', 'longitude']),
    remainder='passthrough'
)

scaled_coordinates = preprocessor.fit_transform(coordinates_cleaned)

# Initialize DBSCAN model
# eps (epsilon): The maximum distance between two samples for one to be considered as in the neighborhood of the other.
# min_samples: The number of samples (or total weight) in a neighborhood for a point to be considered as a core point.
# Adjust these values for optimal results based on data characteristics.
dbscan = DBSCAN(eps=0.05, min_samples=5)

# Fit the model and predict clusters
```

4.4 Visualisasi Cluster GPS (Peta/Scatter Plot)



4.5 Visualisasi Distribusi Cluster

